

```

{
  "creator" : "ari.neko@cvut.cz",
  "timestamp" : "09-07-2026_17:06:54",
  "variant" : 1,
  "subjectName" : "operating-systems",
  "categories" : [ "Deadlocks & Synchronization", "Memory Management", "Kernel Architecture" ],
  "questions" : [ {
    "question" : "What is a binary semaphore commonly called when used strictly as a locking mechanism?",
    "answers" : [ "Condition Variable", "Critical Section", "Mutex", "Monitor" ],
    "correctIndex" : 2
  }, {
    "question" : "What is a critical section inside multi-threaded application software?",
    "answers" : [ "A block of code that accesses shared resources and must not be concurrently run",
      "A critical database constraint tracking primary indexing relationships", "A background process handling system initialization routines", "A protected kernel execution memory register row" ],
    "correctIndex" : 0
  }, {
    "question" : "What classical algorithm can be utilized to safely avoid deadlocks in resource allocation?",
    "answers" : [ "Dijkstra's Shortest Path", "Round Robin Algorithm", "De Morgan's Optimization", "Banker's Algorithm" ],
    "correctIndex" : 3
  }, {
    "question" : "What is the role of a Translation Lookaside Buffer (TLB)?",
    "answers" : [ "To act as a specialized cache for virtual-to-physical address translations", "To map file allocation records sequentially on disk", "To handle network data package buffering streams", "To store execution metrics logs for background tasks" ],
    "correctIndex" : 0
  }, {
    "question" : "What state does a system enter when it spends more time swapping pages than executing instructions?",
    "answers" : [ "Starvation", "Fragmenting", "Paging", "Thrashing" ],
    "correctIndex" : 3
  }, {
    "question" : "What is the main difference between internal and external fragmentation?",
    "answers" : [ "Internal is fixed by defragmentation; external requires virtual memory flushes", "External happens inside kernel space; internal happens in user space", "External happens on hard drives; internal happens inside CPU registers", "Internal occurs inside fixed allocated blocks; external occurs between variable blocks" ],
    "correctIndex" : 3
  }, {

```

```
"question" : "What characterizes a monolithic kernel architecture?",
"answers" : [ "It eliminates context switching overhead entirely", "It runs exclusively on virtual machines", "OS services are moved to separate user-space modules", "All operating system services run inside a single privileged kernel space" ],
"correctIndex" : 3
}, {
"question" : "What is the primary purpose of a system call?",
"answers" : [ "To clear CPU cache configurations automatically", "To compile source code files into machine code", "To manage graphical window rendering pipelines", "To provide an interface for user programs to request services from the kernel" ],
"correctIndex" : 3
}, {
"question" : "What is a major advantage of a microkernel compared to a monolithic design?",
"answers" : [ "Smaller total storage size footprint", "Higher system stability and isolation of corrupted drivers", "Faster execution performance speeds", "Lower context switching overhead costs" ],
"correctIndex" : 1
}, {
"question" : "What privilege layer does the operating system kernel run in on modern CPUs?",
"answers" : [ "Kernel mode (Ring 0)", "Safe mode", "User mode (Ring 3)", "Hypervisor mode" ],
"correctIndex" : 0
}]
}
```